CONICS

GENERAL EQUATION OF SECOND DEGREE

Coordina
Geome
IIT/JEE
Ad

$$ax^2+2hxy+by^2+2gx+2fy+c=0$$

(1) Represents a pair of straight lines if

$$D = abc+2fgh-af^2-bg^2-ch^2=0$$

Lines are parallel if $h^2-ab=0$

Lines are perpendicular if $a+b=0$

Lines are coincidence if $h^2-ab=0$, $g^2-ac=0$ & $f^2-bc=0$

$D = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$

$\Rightarrow$ (1) Circle $a = b$ and $h = 0$

(2) Parabola $h^2 - ab = 0$

(3) Ellipse $h^2 - ab < 0$

(4) Hyperbola $h^2 - ab > 0$

Rectangular Hyperbola $h^2 - ab > 0$

& $a + b = 0$

$$x^2 - 2x + 2y^2 - 8y + 7 = 0$$

Comparing with

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$a = 1$, $b = 2$, $c = 7$

$2h = 0$ $\quad$ $2f = -8$ $\quad$ $2g = -2$

$\Rightarrow h = 0$ $\quad$ $\Rightarrow f = -4$ $\quad$ $\Rightarrow g = -1$

$h^2 - ab$

$\Rightarrow 0^2 - 1 \times 2$

$\Rightarrow -2$

Since $h^2 - ab < 0$, and $a \neq b$, it is an ellipse.